

Assignment8

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1 Revised Cyclist Simulation

Although we have previously simulated a cyclist's velocity through time, it proved to be lacking in realism due to many unaccounted forces. In an effort to remedy this, a drag force was added to the simulation. As it can be seen in Fig.1, the inclusion of the force due to drag gives our cyclist a terminal velocity around 13 m/s. To further improve the simulation, stokes term was also accounted for. Fig.2 shows the negligible affects it has on our cyclist. Finally, the ability to account for slopes was added to the simulation. Fig.3 shows the 33% decrease in the maximum velocity our tireless cyclist can reach due to a 10% grade.

2 Two Random Walkers

In order to simulate two walkers taking random steps of distance 1 forward or backward, for 100 steps, the random module in python was used. Fig.4 shows a simulation of the above situation.

3 Many Walkers

To acquire some statistical data on a large sample of walkers, the code was modified to run 500 at once and record their positions at every step. This data was put into arrays and numpy's built in mean function was used to acquire the mean and mean squared of the displacement at every step. We would predict that the mean remains around zero with some variance to be expected, while the mean squared should increase relatively linearly with a slope of one and show us how far from zero the average walker is. Fig.5 does show the linear trend, however it is not as close to one as may be expected and it seems as if there was a slight tendency to stick closer to zero such that the mean squared is less than one thousand. On the other hand, the mean behaved exactly as expected, with the final mean displacement for this trial being .818 steps. This seems interesting to me as it shows that the walkers that went negative and those that went positive both had a tendency to stick closer to zero as seen by the mean squared being 700 and the mean still being around zero.

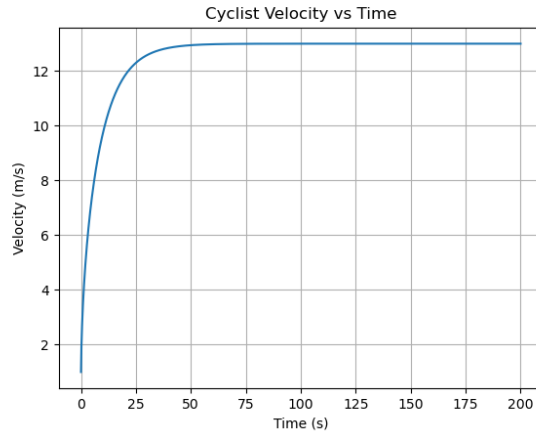


Figure 1: Plot of a cyclist's Velocity Vs Time with drag.

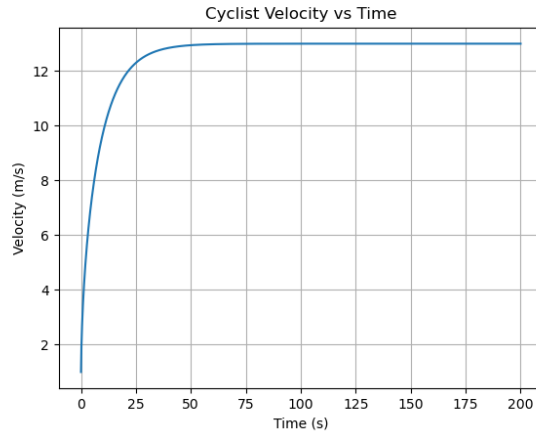


Figure 2: Plot of a cyclist's Velocity Vs Time with drag including stokes term.

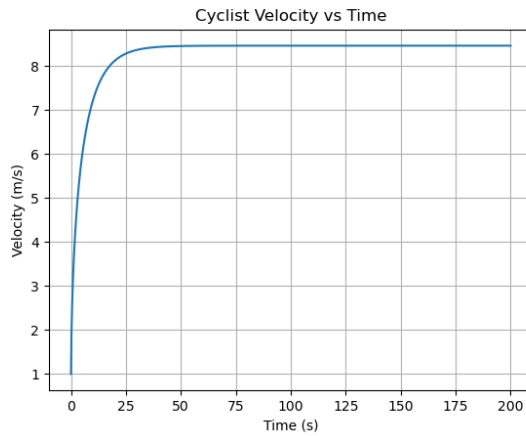


Figure 3: Plot of a cyclist's Velocity Vs Time with drag (including stokes term) and gravity on a 10% grade.

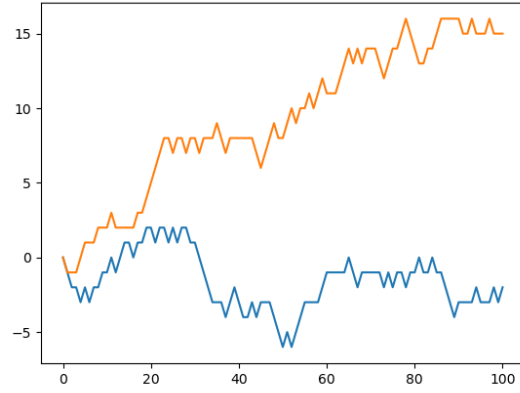


Figure 4: Plot of two random walkers taking 100 random steps.

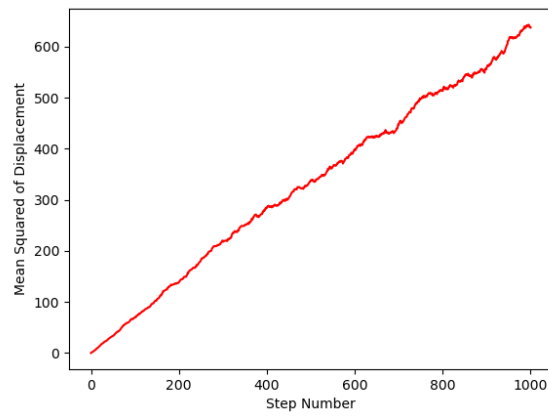


Figure 5: Mean squared of displacement Vs Step number for 500 walkers and 1000 steps each.